

A Project Report
On
Learning Management System

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Under the guidance of
(SP Acharya Sir & Adil Ahmad Sir)

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EVOLVING THINKING MINDS

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CERTIFICATE

This is to certify that Mr. Sharma Abhishek of B.TECH Semester IV has completed her project work titled Learning Management System satisfactorily in partial fulfillment of the requirement of B.TECH degree, Rai University, Saroda, Ahmedabad in the year 2025.

Internal Guide

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DATE : - 11-04-25

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CANDIDATE'S DECLARATION

I hereby declare that, the project report titled “(Project Definition)” submitted towards the partial completion of Project in Semester IV, of the B.TECH in RSE, Rai University, Ahmedabad is an authentic record of my work carried out

I further declare that the work reported in this project has not been submitted and will not be submitted, either in part or in full for the award of any other degree in this institute or any other institute

Candidate's Signature

Candidate's Name : Sharma Abhishek

Branch : B-Tech CSE

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

Signature of Internal Guide

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Current Features

User Management System

Authentication & Authorization

- **Registration Process:**
 - Collects full name, email, password, and phone number
 - Validates input fields with specific length requirements
 - Uploads and processes user avatar using Cloudinary
 - Generates JWT token upon successful registration
 - Implements email verification system
- **Email Verification:**
 - Generates unique verification tokens using crypto
 - Sends verification emails with expiry (15 minutes)
 - Provides resend verification functionality
 - Prevents unverified users from logging in
- **Login System:**
 - Validates credentials against database
 - Uses bcrypt for password comparison
 - Sets HTTP-only cookies with JWT tokens
 - Implements secure cookie options (sameSite, secure flags)
- **Password Management:**
 - Change password functionality for logged-in users
 - Forgot password flow with email-based reset tokens
 - Reset password with token validation
 - Password hashing using bcrypt with pre-save hooks
- **Role-Based Access Control:**
 - Two distinct roles: USER and ADMIN

- Middleware for route protection (isLoggedIn, authorisedRoles)
- Subscription-based access control (authorizeSubscriber)
- Frontend route protection with RequireAuth component

Course Management System

Course Administration

- **Course Creation:**
 - Title, description, category, and creator information
 - Thumbnail upload with Cloudinary integration
 - Validation for field lengths and required fields
 - Unique title constraint
- **Course Modification:**
 - Update course details and thumbnail
 - Delete course functionality with associated content cleanup
 - Admin-only access to modification features
- **Course Structure:**
 - Organized by lectures within courses
 - Metadata tracking (number of lectures, creation date)
 - Category-based organization

Lecture Management

- **Lecture Content:**
 - Title and description fields
 - Video upload capability (up to 500MB)
 - External link support
 - Supplementary materials (PDF) upload
- **Assessment Integration:**
 - Question creation within lectures
 - Multiple-choice question format
 - Correct answer tracking

- Question grouping by lecture
- **Media Handling:**
 - Video hosting via Cloudinary
 - PDF document storage
 - Custom file validation (extensions: jpg, jpeg, webp, png, mp4, pdf)
 - File size limitations and error handling

Subscription & Payment System

Razorpay Integration

- **Payment Processing:**
 - Razorpay API key management
 - Subscription plan configuration
 - Client-side payment initialization
 - Server-side payment verification
- **Subscription Management:**
 - Purchase subscription functionality
 - Subscription verification with cryptographic signature
 - Subscription status tracking
 - Subscription cancellation
- **Payment Records:**
 - Storage of payment IDs and signatures
 - Payment verification status
 - Timestamp tracking
 - Admin access to payment history

Access Control

- **Content Restriction:**
 - Lecture access limited to subscribers and admins
 - Route protection based on subscription status
 - Frontend conditional rendering based on subscription

- API-level checks for subscription validity

Blog System

Blog Administration

- **Blog Creation:**
 - Title, description, and link fields
 - Thumbnail image upload and processing
 - Admin-only creation rights
 - Cloudinary integration for image storage
- **Blog Management:**
 - Listing all blogs with pagination
 - Individual blog retrieval by ID
 - Blog deletion functionality
 - Image optimization and storage

Blog Display

- **Frontend Integration:**
 - Blog listing page with thumbnails
 - Individual blog viewing
 - Responsive design for various devices
 - Image optimization for performance

Analytics & Dashboard

Admin Dashboard

- **User Analytics:**
 - Total registered users count
 - Subscribed users percentage
 - Visual representation using charts
 - Filterable user data
- **Sales Analytics:**
 - Monthly sales records

- Revenue tracking
- Payment success rate
- Visualization with bar and pie charts
- **Course Analytics:**
 - Course popularity metrics
 - Enrollment statistics
 - Category distribution
 - Content growth tracking

User Dashboard

- **Personal Analytics:**
 - Course progress tracking
 - Subscription status display
 - Payment history
 - Profile completion indicators

Future Implementation Roadmap

AI-Enhanced Learning Features

Intelligent Content Recommendation

- **Personalized Learning Paths:**
 - AI algorithms to analyze user learning patterns
 - Course recommendations based on user progress and interests
 - Adaptive difficulty levels based on performance
 - Implementation using collaborative filtering and content-based recommendation systems
- **Smart Content Sequencing:**
 - ML models to determine optimal learning sequences
 - Prerequisite analysis for course dependencies
 - Knowledge gap identification
 - Spaced repetition scheduling for better retention

AI-Powered Assessment System

- **Automated Question Generation:**
 - NLP models to generate questions from lecture content
 - Difficulty calibration based on Bloom's taxonomy
 - Question variety (MCQ, short answer, essay)
 - Implementation using transformer-based language models
- **Intelligent Grading:**
 - Automated essay and short answer grading
 - Plagiarism detection using semantic similarity
 - Code assessment for programming courses
 - Feedback generation with improvement suggestions

Virtual Learning Assistants

- **AI Tutors:**
 - Course-specific chatbots trained on lecture content
 - 24/7 availability for student questions
 - Contextual understanding of course material
 - Implementation using fine-tuned LLMs like GPT-4
- **Study Session Optimization:**
 - AI-driven study scheduling
 - Focus time recommendations based on user productivity patterns
 - Learning style adaptation
 - Integration with calendar systems

Content Enhancement

- **Automated Content Summarization:**
 - AI-generated lecture summaries
 - Key point extraction from videos
 - Concept maps generation
 - Implementation using extractive and abstractive summarization models

- **Multilingual Support:**

- Automated translation of course materials
- Real-time caption generation for videos
- Language preference adaptation
- Implementation using neural machine translation models

Advanced Analytics and Insights

Predictive Analytics

- **Student Success Prediction:**

- Early warning systems for at-risk students
- Completion likelihood models
- Engagement prediction
- Implementation using gradient boosting models

- **Content Effectiveness Analysis:**

- Automated identification of difficult concepts
- Content engagement metrics
- Learning outcome correlation
- A/B testing framework for content variations

Learning Behavior Analysis

- **Engagement Pattern Recognition:**

- Video watching patterns analysis
- Reading time optimization
- Quiz attempt pattern analysis
- Implementation using time-series analysis and clustering

- **Knowledge Retention Tracking:**

- Forgetting curve modeling for individual students
- Spaced repetition optimization
- Long-term retention assessment
- Implementation using Bayesian knowledge tracing

Interactive Learning Experiences

Immersive Learning

- **AR/VR Integration:**
 - 3D model visualization for complex concepts
 - Virtual labs for practical experience
 - Immersive historical or geographical experiences
 - WebXR API integration for browser-based AR/VR
- **Interactive Simulations:**
 - Physics and chemistry experiment simulations
 - Business scenario simulations
 - Medical procedure simulations
 - Implementation using WebGL and Three.js

Collaborative Learning

- **AI-Facilitated Group Formation:**
 - Optimal team composition algorithms
 - Skill complementarity analysis
 - Personality compatibility matching
 - Implementation using clustering and graph algorithms
- **Real-time Collaboration Tools:**
 - Shared document editing with AI suggestions
 - Virtual whiteboards with intelligent object recognition
 - Code pair programming with AI assistance
 - WebRTC and WebSocket implementation

Accessibility and Inclusion

Universal Design for Learning

- **Adaptive Content Presentation:**
 - Automatic format conversion (text-to-speech, video-to-text)
 - Reading level adjustment

- Visual contrast optimization
- Implementation using media processing libraries
- **Cognitive Load Optimization:**
 - Content chunking based on complexity
 - Attention span modeling
 - Cognitive fatigue detection
 - Implementation using user interaction analysis

Neurodiversity Support

- **Learning Style Adaptation:**
 - Visual, auditory, reading/writing, kinesthetic (VARK) preference detection
 - Content presentation adaptation
 - Alternative assessment methods
 - Implementation using preference learning algorithms
- **Focus Assistance:**
 - Distraction reduction interfaces
 - Progress visualization
 - Micro-goal setting
 - Implementation using UX design principles and gamification

Enterprise and Scalability Features

Advanced Content Management

- **AI-Powered Content Auditing:**
 - Automatic outdated content detection
 - Broken link and resource checking
 - Accessibility compliance verification
 - Implementation using web crawling and content analysis
- **Intelligent Content Versioning:**
 - Automatic change detection and highlighting
 - Version comparison with semantic understanding

- Collaborative editing conflict resolution
- Implementation using diff algorithms and version control systems

Integration Ecosystem

- **LTI/xAPI Compliance:**

- Learning Tools Interoperability support
- Experience API for learning activity tracking
- Cross-platform data exchange
- Implementation using standard protocols and APIs

- **Enterprise Authentication:**

- SAML/LDAP integration
- Single Sign-On (SSO) support
- Multi-factor authentication
- Implementation using OAuth 2.0 and OpenID Connect

Mobile and Offline Learning

Progressive Web App Enhancement

- **Offline Content Access:**

- Smart content caching strategies
- Sync mechanisms for offline progress
- Bandwidth optimization
- Implementation using Service Workers and IndexedDB

- **Mobile-First Experience:**

- Touch-optimized interfaces
- Device capability adaptation
- Battery and data usage optimization
- Implementation using responsive design and device APIs

Technical Implementation Roadmap

AI Infrastructure

- **Model Deployment Architecture:**

- Containerized ML model serving
- Model versioning and A/B testing
- Inference optimization for low latency
- Implementation using TensorFlow Serving or PyTorch TorchServe
- **Data Pipeline:**
 - Learning event collection and processing
 - Feature extraction and transformation
 - Training data preparation
 - Implementation using Apache Kafka and Apache Spark

Security Enhancements

- **AI-Based Threat Detection:**
 - Anomalous behavior detection
 - Automated security scanning
 - Fraud prevention for payments
 - Implementation using anomaly detection algorithms
- **Enhanced Privacy:**
 - Federated learning for privacy-preserving analytics
 - Differential privacy for data sharing
 - Granular permission controls
 - Implementation using privacy-preserving ML techniques

Performance Optimization

- **Intelligent Caching:**
 - Predictive content preloading
 - User-specific cache prioritization
 - Cache invalidation strategies
 - Implementation using Redis and client-side storage
- **Adaptive Resource Allocation:**
 - Dynamic scaling based on usage patterns

- Resource optimization for cost efficiency
- Load balancing with traffic prediction
- Implementation using Kubernetes and cloud autoscaling

Implementation Timeline

Phase 1: Foundation AI Features (6 months)

- Basic recommendation system
- Automated quiz generation from content
- Simple chatbot assistance
- Learning analytics dashboard

Phase 2: Enhanced Learning Experience (6-12 months)

- Personalized learning paths
- Advanced content recommendations
- Improved virtual assistants
- Predictive analytics for student success

Phase 3: Advanced AI Integration (12-18 months)

- Immersive AR/VR experiences
- Sophisticated assessment systems
- Adaptive content presentation
- Enterprise integration features

Phase 4: Ecosystem Expansion (18-24 months)

- Full AI tutor capabilities
- Cross-platform learning experiences
- Advanced accessibility features
- Comprehensive analytics ecosystem

Technical Requirements

Infrastructure

- Cloud-based ML model hosting
- GPU resources for model training

- Scalable database architecture
- Content delivery network optimization

Expertise

- Machine learning engineers
- Natural language processing specialists
- Educational technology experts
- UX researchers for learning experience

Data Requirements

- Learning interaction data collection
- Content metadata standardization
- User preference and performance tracking
- Privacy-compliant data governance

All routes :

Backend API Routes

User Management Routes (/api/v1/user)

HTTP Method	Endpoint	Controller	Function Description	Authentication Required	Authorization Required
POST	/register	register	User registration with avatar upload	No	No
POST	/login	login	User authentication	No	No
GET	/logout	logout	User logout	No	No
GET	/getuser	getAllUsers	Retrieve all users	No	No

HTTP Method	Endpoint	Controller Function	Description	Authentication Required	Authorization Required
POST	/updateuser/:userId	updateUserByAdmin	Admin updates user details	No	No
POST	/verify-email/:token	verifyEmail	Email verification	No	No
POST	/resend-verification	resendVerificationEmail	Resend verification email	No	No
GET	/getUserById/:id	getUserById	Get specific user by ID	No	No
GET	/me	getProfile	Get current user profile	Yes	No
POST	/reset	forgotPassword	Initiate password reset	No	No
POST	/reset/:resetToken	resetPassword	Reset password with token	No	No
POST	/change-password	changePassword	Change password	Yes	No
POST	/update/:id	updateUser	Update user profile	Yes	No

Course Management Routes (/api/v1/courses)

HTTP Method	Endpoint	Controller Function	Description	Authentication Required	Authorization Required
GET	/	getAllCourses	Get all courses	No	No

HTTP Method	Endpoint	Controller Function	Description	Authentication Required	Authorization Required
POST	/	createCourse	Create new course	Yes	ADMIN
DELETE	/	deleteCourseLecture	Delete lecture from course	Yes	ADMIN
PUT	/	updateCourseLecture	Update lecture in course	Yes	ADMIN
GET	/:id	getLecturesByCourseId	Get course lectures	Yes	Subscriber
PUT	/:id	updateCourse	Update course details	Yes	ADMIN
DELETE	/:id	removeCourse	Delete course	Yes	ADMIN
POST	/:id	addLectureToCourseById	Add lecture to course	Yes	ADMIN

Payment Routes (/api/v1/payments)

HTTP Method	Endpoint	Controller Function	Description	Authentication Required	Authorization Required
GET	/razorpay-key	getRazorPayApiKey	Get Razorpay API key	Yes	No
POST	/subscribe	buySubscription	Purchase subscription	Yes	No
POST	/verify	verifySubscription	Verify subscription payment	Yes	No

HTTP Method	Endpoint	Controller Function	Description	Authentication Required	Authorization Required
POST	/unsubscribe	cancelSubscription	Cancel subscription	Yes	No
GET	/	allPayments	Get all payment records	Yes	ADMIN

Blog Routes (/api/v1/blog)

HTTP Method	Endpoint	Controller Function	Description	Authentication Required	Authorization Required
POST	/create	createBlog	Create new blog post	No	No
GET	/all	getAllBlogs	Get all blog posts	No	No
GET	/:id	getBlogById	Get specific blog post	No	No
DELETE	/:id	deleteBlog	Delete blog post	No	No

Miscellaneous Routes (/api/v1/)

HTTP Method	Endpoint	Controller Function	Description	Authentication Required	Authorization Required
POST	/contact	contactUs	Submit contact form	No	No
GET	/admin/stats/users	stats	Get user statistics	Yes	ADMIN

Frontend Routes

Public Routes

Route Path	Component	Description	Protected
/	HomePage	Landing page	No

Route Path	Component	Description	Protected
/about	AboutUs	About page	No
/contact	Contact	Contact form	No
/denied	Denied	Access denied page	No
/blog	Blog	Blog listing page	No
/verify-email/:token	EmailVerification	Email verification	No
/resend-verification	ResendVerification	Resend verification email	No
/signup	Signup	User registration	No
/login	Login	User login	No
/user/profile/reset-password	ForgotPassword	Forgot password form	No
/user/profile/reset-password/:resetToken	ResetPassword	Reset password with token	No
/courses	CourseList	Course catalog	No
/courses/description	CourseDescription	Course details	No
*	NotFound	404 page	No

Admin-Only Routes

Route Path	Component	Description	Protected
/course/create	CreateCourse	Create new course	Yes (ADMIN)
/course/addlecture	AddLecture	Add lecture to course	Yes (ADMIN)
/admin/dashboard	AdminDashboard	Admin dashboard	Yes (ADMIN)
/admin/userdata	Userdata	User management	Yes (ADMIN)
/admin/addblog	AddBlog	Create blog post	Yes (ADMIN)
/admin/users/:userId/edit	AdminUserEdit	Edit user details	Yes (ADMIN)

User & Admin Routes

Route Path	Component	Description	Protected
/user/profile	Profile	User profile	Yes (USER, ADMIN)
/user/profile/change-password	ChangePassword	Change password	Yes (USER, ADMIN)
/checkout	Checkout	Subscription checkout	Yes (USER, ADMIN)
/checkout/success	CheckoutSuccess	Payment success	Yes (USER, ADMIN)
/checkout/fail	CheckoutFail	Payment failure	Yes (USER, ADMIN)
/course/displaylectures	DisplayLecture	View course lectures	Yes (USER, ADMIN)

API Integration Details

Authentication Flow

1. Registration Process:

- Frontend: User submits form at /signup
- Backend: POST request to /api/v1/user/register
- Response: JWT token set in HTTP-only cookie
- Redirect: User directed to email verification notice

2. Email Verification:

- Frontend: User clicks link in email to /verify-email/:token
- Backend: POST request to /api/v1/user/verify-email/:token
- Response: Verification status
- Redirect: User directed to login page on success

3. Login Process:

- Frontend: User submits credentials at /login
- Backend: POST request to /api/v1/user/login
- Response: JWT token set in HTTP-only cookie

- Redirect: User directed to homepage or dashboard based on role

4. Password Reset:

- Frontend: User requests reset at /user/profile/reset-password
- Backend: POST request to /api/v1/user/reset
- Email: Reset link sent to user
- Frontend: User sets new password at /user/profile/reset-password/:resetToken
- Backend: POST request to /api/v1/user/reset/:resetToken

Course Management Flow

1. Course Creation:

- Frontend: Admin submits form at /course/create
- Backend: POST request to /api/v1/courses
- Response: Course details with ID
- Redirect: Admin directed to lecture addition page

2. Lecture Addition:

- Frontend: Admin uploads content at /course/addlecture
- Backend: POST request to /api/v1/courses/:id
- Response: Updated course with new lecture
- State Update: Course lectures list refreshed

3. Course Viewing:

- Frontend: User browses courses at /courses
- Backend: GET request to /api/v1/courses
- Frontend: User selects course to view lectures at /course/displaylectures
- Backend: GET request to /api/v1/courses/:id
- Conditional: Access granted only to subscribers or admins

Payment Integration

1. Subscription Purchase:

- Frontend: User initiates at /checkout

- Backend: GET request to /api/v1/payments/razorpay-key
- Backend: POST request to /api/v1/payments/subscribe
- Frontend: Razorpay payment form displayed
- User Action: Payment completed
- Backend: POST request to /api/v1/payments/verify
- Redirect: User directed to /checkout/success or /checkout/fail

2. Subscription Cancellation:

- Frontend: User cancels from profile page
- Backend: POST request to /api/v1/payments/unsubscribe
- Response: Updated subscription status
- State Update: User subscription status updated

Blog Management Flow

1. Blog Creation:

- Frontend: Admin creates at /admin/addblog
- Backend: POST request to /api/v1/blog/create with multipart form data
- Response: Blog details with ID
- State Update: Blog list refreshed

2. Blog Viewing:

- Frontend: Users view at /blog
- Backend: GET request to /api/v1/blog/all
- Frontend: Individual blog viewing
- Backend: GET request to /api/v1/blog/:id (if implemented)

Admin Dashboard Flow

1. Statistics Viewing:

- Frontend: Admin views at /admin/dashboard
- Backend: GET request to /api/v1/admin/stats/users
- Response: User statistics data
- Frontend: Data visualization rendered

2. User Management:

- Frontend: Admin views at /admin/userdata
- Backend: GET request to /api/v1/user/getuser
- Frontend: Admin edits user at /admin/users/:userId/edit
- Backend: POST request to /api/v1/user/updateuser/:userId

Route Protection Mechanisms

Backend Protection

1. Authentication Middleware (isLoggedIn):

- Verifies JWT token from cookies
- Sets user details in request object
- Returns 400 error if not authenticated

2. Role Authorization Middleware (authorisedRoles):

- Checks user role against allowed roles
- Returns 403 error if unauthorized
- Allows access if role matches

3. Subscription Verification Middleware (authorizeSubscriber):

- Checks user subscription status
- Allows ADMIN role to bypass
- Returns 403 error if not subscribed

Frontend Protection

1. RequireAuth Component:

- Wraps protected routes in <RequireAuth> component
- Checks user authentication status from Redux store
- Redirects to login if not authenticated
- Verifies user role against allowed roles
- Redirects to denied page if unauthorized

2. Conditional Rendering:

- Admin-only UI elements shown based on role

- Subscription-required features conditionally displayed
- Payment options shown based on subscription status

Data Flow Architecture

Redux Store Structure

1. Auth Slice:

- Manages user authentication state
- Stores user details, role, and login status
- Handles login, logout, and registration actions

2. Course Slice:

- Stores course catalog data
- Manages course creation and update actions
- Tracks course loading and error states

3. Lecture Slice:

- Manages lecture data for selected course
- Handles lecture addition and deletion
- Tracks lecture loading and error states

4. Razorpay Slice:

- Stores payment API key
- Manages subscription status
- Tracks payment processing state

5. Stat Slice:

- Stores admin dashboard statistics
- Tracks data loading and error states

6. Blog Slice:

- Manages blog post data
- Handles blog creation and retrieval
- Tracks blog loading and error states

API Communication

1. Axios Instance:

- Centralized API configuration
- Automatic token inclusion in requests
- Response interceptors for error handling

2. Toast Notifications:

- Success and error feedback for API operations
- Loading indicators during async operations
- Consistent user feedback pattern

Security Considerations

1. Authentication Security:

- HTTP-only cookies for JWT storage
- CORS configuration with specific origins
- Secure and SameSite cookie settings

2. Input Validation:

- Server-side validation for all inputs
- File upload restrictions and validation
- MongoDB schema validation

3. Authorization Controls:

- Fine-grained role-based access
- Subscription status verification
- Resource ownership verification

4. Error Handling:

- Custom error middleware
- Sanitized error responses
- Consistent error format

Screenshots: